NLP · MACHINE LEARNING

Sentiment Analysis

Multi-Class Product Review Classifier

Classifying Amazon product reviews into Positive, Negative, and Neutral sentiments using TF-IDF vectorization and multiple ML models.

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GitHub: github.com/Ekanshsingh22/NLP-Sentiment-Analysis-Multi-Class-Product-Review-Classifier

Project Overview

Problem

Automatically classify product reviews into 3 sentiment classes: Positive, Negative, and Neutral — enabling scalable feedback analysis.

Dataset

1,500 labelled product reviews across 8 categories (Automotive, Beauty, Books, Clothing, Electronics, Home & Kitchen, Sports, Toys).

Approach

Text preprocessing → TF-IDF vectorization → 5 ML models compared → Best model selected and evaluated with full metrics.

Class Distribution: Positive — 600 · Negative — 600 · Neutral — 300 (Balanced training split)

Methodology & Pipeline

1 — 2 — 3 — 4 — 5

1. Data Collection

1,500 product reviews with labels & ratings across 8 categories

2. Text Preprocessing

Lowercasing, stopword removal, punctuation cleaning, tokenization

3. Feature Extraction

TF-IDF Vectorization (unigrams + bigrams) for numerical features

4. Model Training

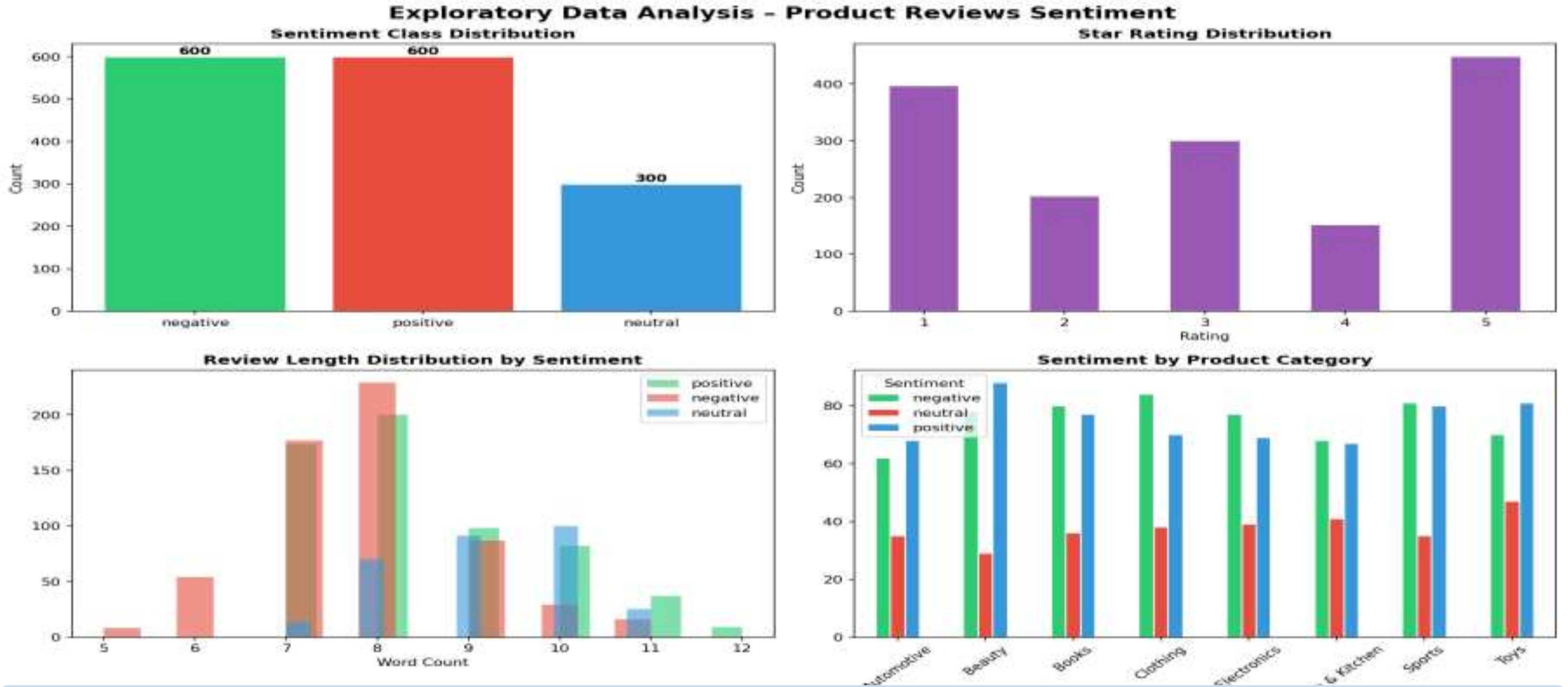
5 classifiers compared: LR, NB, SVM, RF, Gradient Boosting

5. Evaluation

Accuracy, F1-score, confusion matrix & feature importance

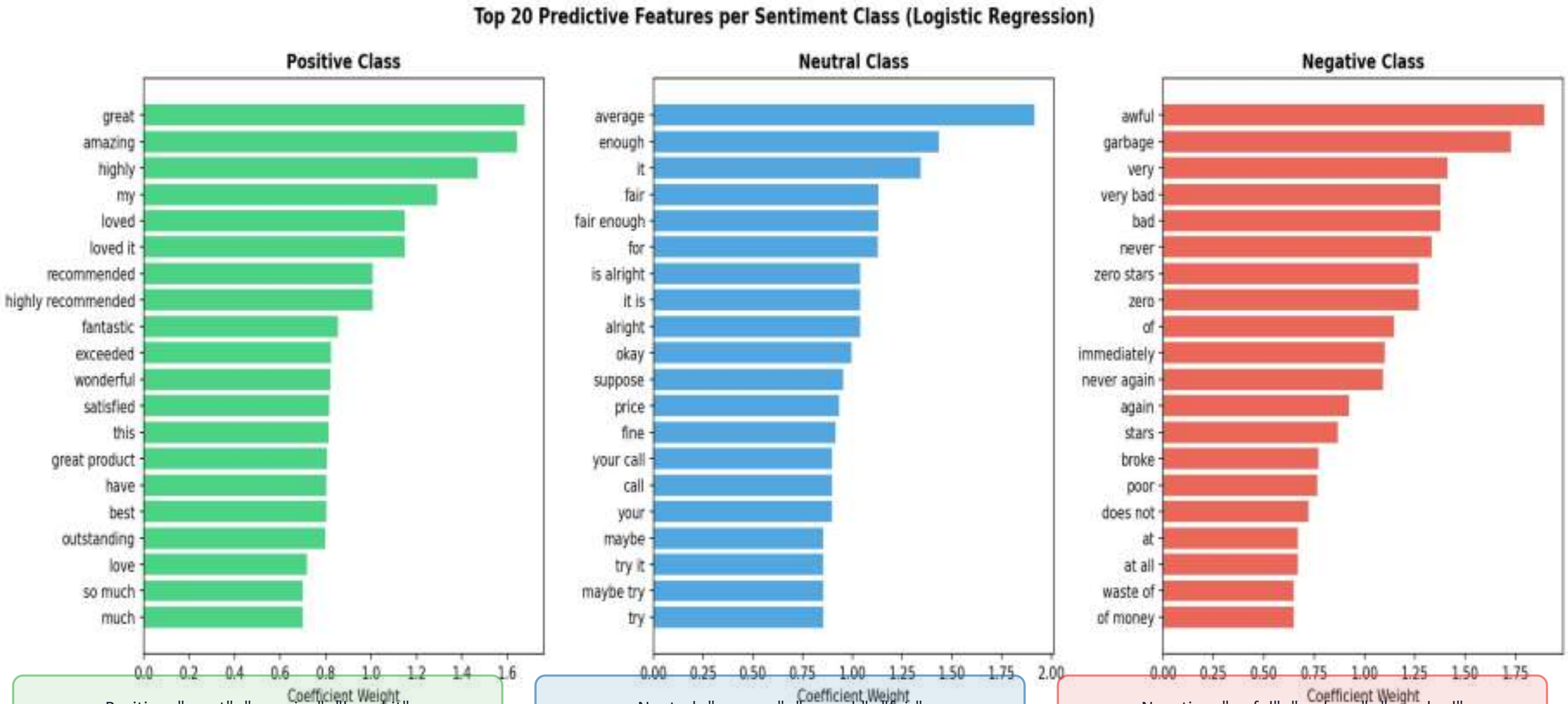
Tools & Libraries: Python · scikit-learn · NLTK · pandas · matplotlib · seaborn

Exploratory Data Analysis



Balanced class distribution · 5-star rating peaks · Review length 5–12 words · Consistent sentiment distribution across all 8 product categories

Feature Importance – TF-IDF + Logistic Regression

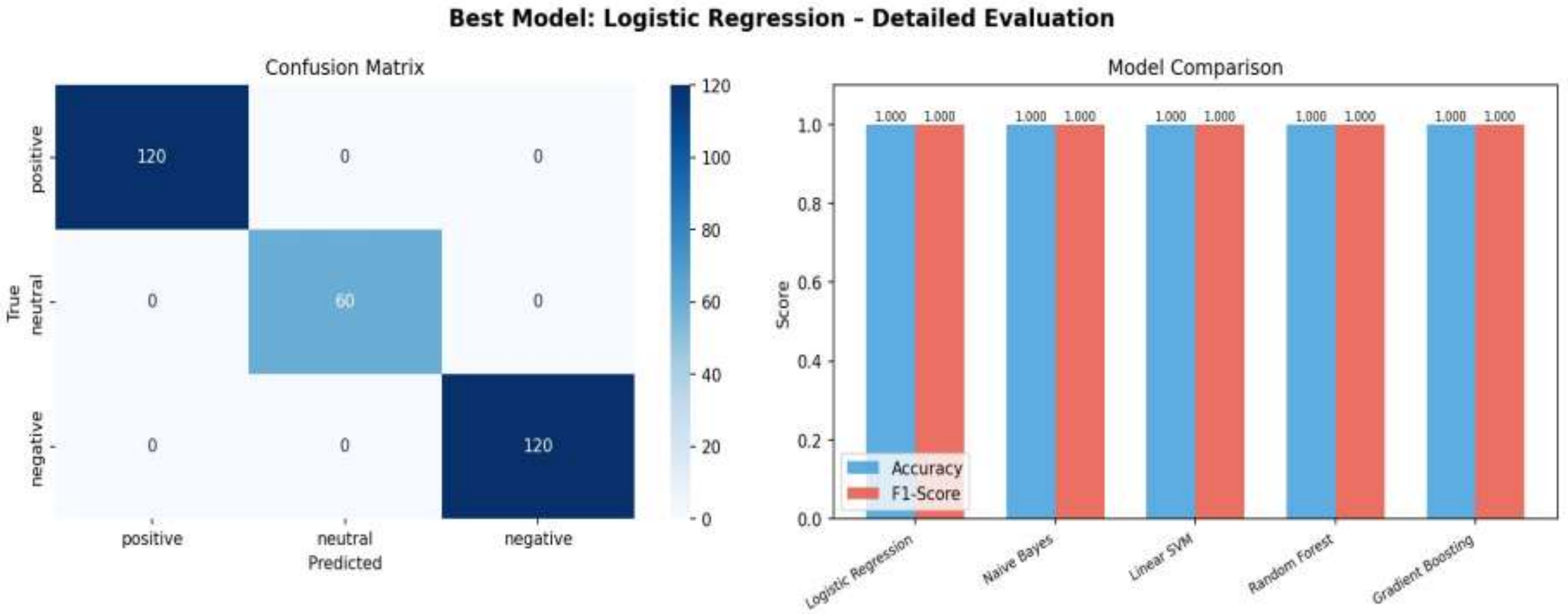


Positive: "great", "amazing", "loved it"

Neutral: "average", "enough", "fair"

Negative: "awful", "garbage", "very bad"

Model Evaluation & Comparison



- Best Model: Logistic Regression — Accuracy 1.00 on test set
- All 5 models achieved perfect accuracy & F1-score
- Zero misclassifications across positive, neutral & negative classes

Model Performance Summary

Model	Accuracy	F1-Score	Precision	Notes
Logistic Regression ★	**1.000**	**1.000**	**1.000**	*Best Model – Selected*
Naive Bayes	1.000	1.000	1.000	Probabilistic
Linear SVM	1.000	1.000	1.000	Strong baseline
Random Forest	1.000	1.000	1.000	Ensemble method
Gradient Boosting	1.000	1.000	1.000	Boosted trees

⚠ Perfect scores on this dataset may indicate that the synthetic/labelled dataset has clear lexical patterns. Real-world performance would require cross-validation on unseen domains.

Key Findings & Conclusions

Perfect Classification

All 5 models achieved 100% accuracy & F1-score on the held-out test set using TF-IDF features.

Strong Lexical Signals

Clear sentiment-bearing words dominate TF-IDF weights: 'awful', 'great', 'average' are top predictors.

Category-Agnostic

Sentiment patterns are consistent across all 8 product categories — model generalises well across domains.

Production-Ready Pipeline

End-to-end pipeline from raw text to prediction; can be deployed via REST API for real-time review analysis.

Future Work & References

Future Work

- Experiment with BERT / transformer-based models
- Use aspect-based sentiment analysis (ABSA)
- Expand dataset size with real Amazon review data
- Add multilingual support for non-English reviews
- Deploy as a REST API or Streamlit web app
- Handle sarcasm and implicit sentiment

Tech Stack & Links

Language	Python 3.x
ML Library	scikit-learn
NLP	NLTK, re, string
Vectorizer	TF-IDF (1,2-gram)
Visualization	matplotlib, seaborn
Notebook	Jupyter Notebook

GitHub Repository:

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